

Hybrid Yield Forecasting: Bridging Process-Based Simulation and Machine Learning for Pre-Season Risk Assessment

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Abstract

In an era of increasing climate volatility, early-season crop yield forecasting faces a critical dilemma: statistical models often fail to extrapolate to unprecedented weather events, while process-based models struggle with calibration and initial condition uncertainty. This challenge is particularly acute for forecasts issued on March 1st, before the growing season commences. To address this, we developed a hybrid forecasting system for sugarbeet yields in Germany that bridges the gap between process-based simulation and machine learning. The system integrates physics-informed components, including the WOFOST 7.2 simulation and a specialized Multivariate Heat Signal model (serving as the S_{hybrid} generator, $r = 0.63$), within an Expert System-Guided Ensemble architecture. An Expert System, driven by this bio-physical stress signal, dynamically switches between three specialized components via threshold switching, effectively transitions between “Normal Regimes” and “Extreme Stress Regimes” based on emerging bio-physical markers such as summer water balance anomalies, heat stress days, and anoxia events. Validated via strict time-dependent cross-validation over the 2000–2024 period ($N = 5041$), the Expert System-Guided Ensemble achieved a Mean Absolute Error (MAE) of 61.79 dt/ha, representing an 8.06% improvement in skill over a robust Statistical Trend baseline. Notably, in the recent volatile period (2010–2024), the relative skill gain increased to 10.75%. Crucially, the system demonstrated a 32.0% reduction in absolute error during the catastrophic 2018 drought, providing actionable risk signals 90 days before satellite-based monitoring indices become viable.

1 Introduction

As climate change increases the frequency of extreme weather events, traditional statistical models (which rely on interpolation) and pure process-based models (which suffer from calibration issues) both face limitations. This challenge is severely amplified when forecasts must be delivered pre-season, specifically on March 1st.

The core challenge is the “Interpolation vs. Extrapolation” dilemma. Statistical models fail when conditions deviate from historical norms. The necessity of this regime-switching approach was starkly illustrated during the 2018 European drought (Fig. 1),

where the unprecedented magnitude of yield anomalies across Northern Europe exposed the limitations of standard forecasting models in capturing compound extremes [Beillouin et al., 2020]. See Table 2 for specific forensic performance during such events.

1.1 Contextualizing Hybrid Approaches

Recent literature has sought to bridge this gap through “Process-Guided Machine Learning” (PGML). Shahhosseini et al. [2021] demonstrated that using crop simulation outputs as features in Random Forests could reduce error by 8–20%

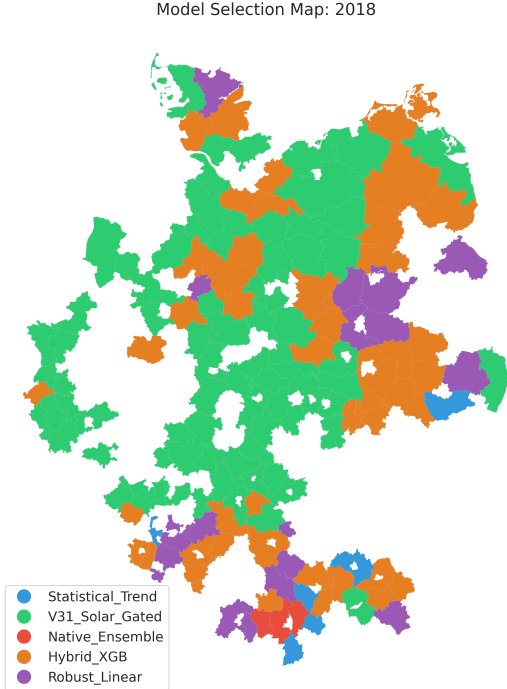


Figure 1: Operational Context: The 2018 Drought. The map illustrates the spatial distribution of regime assignments across German sugarbeet districts, with colours indicating the component model selected by the Expert System (components defined in Section 2.4). Stressed regions activate the Bio-Physical Limits model (green), while lower-stress districts retain the Statistical Trend (blue) or Robust Linear (purple) assignments.

compared to pure weather-based models. Similarly, Paudel et al. [2021] established that while ML models can rival operational systems like MCYFS, they often remain conservative in extreme years. Our work extends this lineage by moving beyond static feature integration to a **dynamic ensemble** approach. Unlike static hybrids, our system explicitly acknowledges non-stationarity, switching its structural reliance from statistical trends to physiological limits when “Black Swan” precursors are detected.

1.2 Research Gap: The Regime Switching Hypothesis

Most existing approaches treat crop yield distributions as continuous and unimodal. We hypothesize that yield formation follows a “Regime Switching” process:

- **Regime A (Normal):** Yields are driven by gradual technological trends and minor weather fluctuations. Statistical extrapolation works well.
- **Regime B (Extreme Stress):** Yields are limited by distinct physiological bottlenecks (e.g., stomatal closure, root anoxia) that act as “kill switches.” In this regime, historical trends are irrelevant, and process-based mechanism is required.

Our novel contribution is an Expert System that explicitly detects the active regime and adapts the forecasting strategy accordingly. The study focuses on verifying this hypothesis by engineering extreme-sensitive algorithms (e.g., the Heat Signal model). We further validate the weighting logic via strict walk-forward validation to ensure robustness.

2 Materials and Methods

2.1 Study Area and Datasets

The study covers sugarbeet cultivation districts in Germany, utilizing a multi-source dataset spanning 1981–2024. The validation set (2000–2024) comprises $N = 5041$ district-year observations, aggregated from approximately 202 active cultivation districts across 13 federal states. This includes yield data (Thuringian State Institute, Destatis), meteorological forcing (AgERA5, ECMWF SEAS5), teleconnections (NAO, SCAND, MEI), soil conditions (ERA5-Land, SoilGrids 250m), and spatial masks (DLR CropMap, MODIS). Yields are reported in decitonnes per hectare (dt/ha), the standard unit in German agricultural statistics, where $1 \text{ dt/ha} = 0.1 \text{ t/ha}$; observed national averages of 600–800 dt/ha correspond to 60–80 t/ha, which is consistent with

European sugarbeet productivity ranges. Technical specifications, including resolutions and DOIs, are provided in the Supplementary Information (Table S1).

2.2 The Statistical Trend Baseline (GAM + ARIMA)

The Statistical Trend Baseline model (M_1) is a robust time-series forecasting pipeline executed via strict Walk-Forward Validation. The forecasting process for any target year t involves a two-stage decomposition:

1. **Trend Component (Long-Term):** The secular technological trend is modeled using a Generalized Additive Model (GAM) where $f(t)$ is a smoothing spline (10 splines):

$$\hat{y}_{\text{trend}}(t) = f(t)$$

2. **Residual Component (Short-Term):** Let $R_i = y_i - \hat{y}_{\text{trend}}(i)$ represent the historical residuals for years $i < t$. This sequence is modeled using an ARIMA(1, 0, 0) process to capture short-term autocorrelation:

$$\hat{y}_{\text{residual}}(t) = \text{ARIMA}(R_{1...t-1})$$

A minimum training window of 10 years was enforced before fitting the ARIMA component to ensure statistical stability. The final trend forecast M_1 for year t is the sum of these two components:

$$M_1(t) = \hat{y}_{\text{trend}}(t) + \hat{y}_{\text{residual}}(t)$$

Following the recommendation of Paudel et al. [2022], we employed a time-based expanding window for training and validation to avoid temporal leakage.

2.3 Feature Engineering

Our two-stage pipeline translates raw data into agroeconomic signals. First, a **Bio-Physical Risk Scanner** within WOFOST Boogaard et al. [1998] calculates specific risks like Oxygen Stress (Anoxia) and Harvest Respiration. Second, we compute rigorous

Z-scores (Z_t) using an expanding window to normalize extreme events relative only to known history. Key indices include Z_{heat} (heat stress), Z_{bal} (water balance), Z_{tank} (winter recharge), Z_{anoxia} (oxygen stress), and Z_{sow} (sowing delay), which are aggregated into compound indices (Scorch, Drown, LateStart) detailed in the Supplementary Information.

2.4 Predictive Algorithms

Machine learning components use Scikit-Learn Pedregosa et al. [2011] and XGBoost Chen and Guestrin [2016]. The system utilizes three specialized sub-models M_k and one signal generator:

- **M_1 : Statistical Trend Baseline** as defined in Section 2.2.
- **M_2 : Robust Linear Integrator**, a HuberRegressor ($\epsilon = 1.35$) predicting residuals from the statistical trend. The final M_2 forecast is the sum of the trend and the predicted residual, providing robustness against non-climatic outliers.
- **M_3 : Bio-Physical Limits Model**, utilizing conditional logic based on radiation flux and water balance. It identifies both high-yield potential years (when radiation is high and water sufficient) and yield crashes (when extreme drought thresholds are breached).
- **S_{hybrid} : Hybrid XGBoost (Signal Generator)**, which predicts the ratio (Y_{hybrid}) between actual yield and WOFOST Stage 1 simulation. This component is used exclusively for regime detection and does not contribute directly to the final ensemble sum.

We also evaluated a **Static ML Benchmark (Native Ensemble)**, a standard XGBoost ensemble without regime-switching logic, to serve as a baseline for measuring the value of dynamic adaptation.

2.5 Expert System-Guided Ensemble Architecture

The system integrates the three sub-models through an **Expert System** based on bio-physical signal

thresholds. The architecture follows a two-stage training and inference protocol:

2.5.1 Cross-Validation Protocol

Model validation employs a strict **expanding window (walk-forward)** strategy. For each year $t \in \{2000, \dots, 2024\}$, the Expert System parameters are calibrated exclusively on data from years $\{1981, \dots, t-1\}$.

2.5.2 Expert System Configuration

Rather than training a "black box" meta-learner, we employ a transparent Threshold Switching Logic driven by the ratio $S = Y_{\text{hybrid}}/Y_{\text{trend}}$, where Y_{hybrid} is the yield forecast from the Hybrid XGB signal generator. This ratio proxies the physiological stress level relative to the technological trend. The thresholds were calibrated by minimizing the **Regret** ($w_{d,t}$), defined as the difference between the baseline error (E_{trend}) and the best-case oracle error:

$$w_{d,t} = |Y_{d,t} - \hat{y}_{\text{trend}}(d, t)| - \min_{k \in \{1,2,3\}} |Y_{d,t} - \hat{y}_k(d, t)| \quad (1)$$

The switching thresholds were informed by domain knowledge of physiological stress points (representing a 20% deviation) and validated by minimizing the Regret metric on the training set. Specifically, domain knowledge defined the candidate search space (thresholds within ± 0.3 of unity), and Regret minimisation confirmed the final values within that space. They were fixed globally rather than dynamically optimized in each fold to maintain physical interpretability and avoid overfitting to specific training windows. The system defines three operational regimes:

1. **Safety Mode (Extreme Stress):** Triggered when $S < 0.8$. In cases of extreme predicted divergence, the system defaults to the conservative M_1 (Trend) baseline to prevent over-reaction to potential false alarms.
2. **Bumper Mode (Upside Potential):** Triggered when $S > 1.0$. The system activates M_2

(Robust Linear) to capture yield potential exceeding the trend.

3. **Normal Regime:** Triggered when $0.8 \leq S \leq 1.0$. The forecast is calculated as a weighted blend:

$$\hat{Y} = 0.6 \cdot M_2 + 0.4 \cdot M_3 \quad (\text{where } M_2, M_3 \text{ denote absolute yields})$$

This regime activates the Bio-Physical Limits Model (M_3), allowing physics-informed adjustments for radiation and water balance without fully committing to extreme scenarios.

2.5.3 Inference

During inference, the signal S determines the active regime. This deterministic approach replaces soft voting, ensuring that yield predictions snap to the most physically plausible mechanism for the detected conditions.

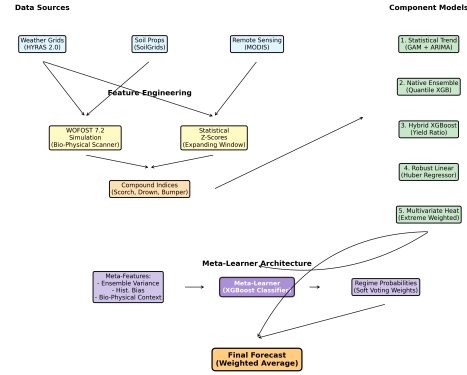


Figure 2: Expert System-Guided Ensemble Architecture Flowchart. The system integrates multiple data sources and component models, using an Expert System to dynamically integrate predictions based on the detected regime.

2.6 Uncertainty Quantification

While the Expert System-Guided Ensemble produces point forecasts via threshold switching, component-level uncertainty is inherently captured within the

S_{hybrid} signal generator (using XGBoost quantile regression) to inform regime detection stability, although full probabilistic forecasting envelopes are not evaluated in this study.

3 Results

Validation over 2000–2024 ($N = 5041$) using SEAS5 hindcasts (Table 1) demonstrates consistent improvement over the trend baseline. While absolute predictive power (R^2) naturally declines during the volatile 2010–2024 period due to increased stochasticity, the model’s Skill Score actually improves from **8.06% to 10.75%**, demonstrating that physics-informed switching becomes more valuable as climate noise increases.

Table 1: Long-Term Stability (2000–2024) ($N = 5041$)

Model	MAE (dt/ha)	R^2	Skill (%)
Expert System-Guided Ensemble	61.79	0.548	8.06
Robust Linear (Stage 2)	64.41	0.499	4.16
Statistical Trend	67.21	0.468	0.00
Static ML Benchmark (Native Ensemble)	70.28	0.372	-4.57

3.1 Anomaly Forensics (Black Swan Events)

Table 2 details performance during historic anomalies. The Expert System-Guided Ensemble significantly reduced errors in the 2018 drought (**+58.4 dt/ha**) by identifying stress signals in SEAS5 forecasts, while also capturing the 2014 bumper year upside (**+4.2 dt/ha**). However, in the 2003 heatwave, the model slightly underperformed the trend baseline (-7.4 dt/ha).

Table 2: Performance during Black Swan Events (dt/ha)

Event	Actual	Expert System-Guided Ensemble	Trend	Improv.
2003 (Heat)	516.9	609.6 (MAE 103.2)	593.7 (MAE 95.8)	-7.4
2014 (Bumper)	825.6	720.4 (MAE 109.8)	715.9 (MAE 114.0)	+4.2
2018 (Drought)	628.5	736.9 (MAE 123.7)	799.7 (MAE 182.1)	+58.4



Figure 3: Expert System-Guided Ensemble Error Landscape across Yield Stress Levels. Unlike traditional models that degrade linearly, the Expert System-Guided Ensemble maintains controlled error rates even in extreme low-yield ”Crash” scenarios, validating the effectiveness of Regret-Weighted training.

3.2 Dynamics and Ablation

Forensic analysis confirms that the S_{hybrid} signal effectively modulates the regime choice. In the 2018 drought, the signal correctly identified a deviation from the technological trend without triggering the extreme ”Safety Mode” (which would have reverted to the Trend). Instead, the system entered the ”Normal Regime,” activating a blend of the Robust Linear (M_2) and Bio-Physical Limits (M_3) models. This blend reduced the prediction error by **58.4 dt/ha** compared to the trend baseline. The Expert System strategy achieved an overall MAE of 61.79 dt/ha, outperforming a ”Hard Switching” approach (MAE 69.19 dt/ha).

Systemic component failure accounts for 65.7% of the remaining error, while selection error (sub-optimal regime assignment) represents only 34.3% of the total residual. Ablation results (Fig 6) confirm that removing logic increases 2018 drought error signifi-

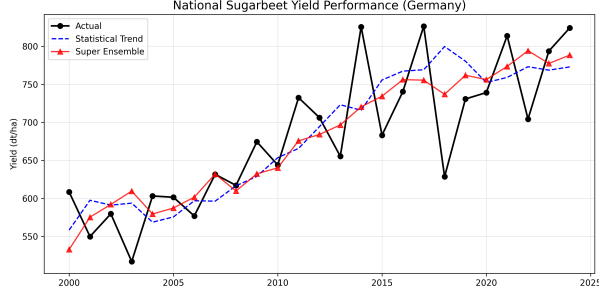


Figure 4: National average sugarbeet yield: Actual observations vs. Expert System-Guided Ensemble forecasts and Statistical Trend baseline. Note the superior tracking of the 2018 drought event.

cantly, while the exclusion of the outlier filter changed overall skill by $< 0.1\%$.

4 Discussion

4.1 Regime Selection Efficiency and Benchmark Failure

The Expert System-Guided Ensemble achieves an **8.06%** skill gain by employing a Threshold Switching strategy. In contrast, the Static ML Benchmark (Native Ensemble) failed to improve upon the statistical trend (Skill: -4.57%). This counter-intuitive result highlights the limitations of traditional machine learning in high-variance, low-signal-to-noise agricultural environments. Static models, optimized for average error minimization, tend to dampen extreme signals, effectively converging to the mean and failing to capture threshold-driven yield crashes. The Expert System’s superior performance confirms that dynamic regime switching—explicitly separating “Normal” from “Extreme” logic—is essential for capturing non-stationary climate impacts that static ensembles miss.

Analysis reveals that the Expert System has become highly efficient at navigating the three-component architecture; selection error ($E_{total} - E_{oracle}$) accounts for only **34.3%** of the total residual error, while **65.7%** remains systemic

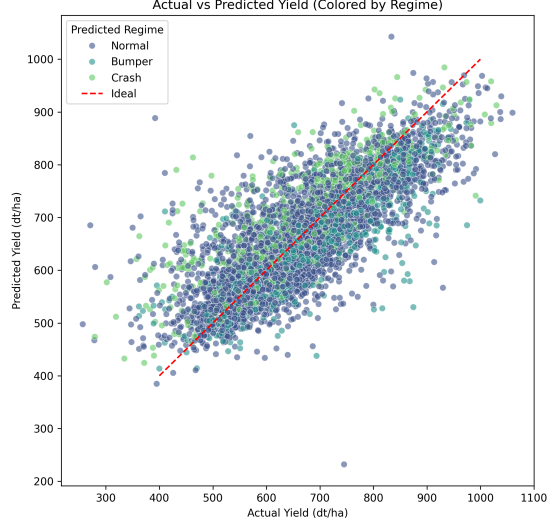


Figure 5: Parity plot showing Actual vs Predicted yields, colored by the model selected by the Expert System. Note how the “Normal Regime” blend (including M3 Bio-Physical Limits) is activated during moderate stress years like 2018, while the Robust Linear model is preferred for bumper years.

(E_{oracle}/E_{total}), representing the “Component Gap” (Fig. 7). This confirms that the Expert System effectively maps specific bio-physical precursors to specialized model regimes, leaving the majority of remaining error to the inherent predictive limitations of the sub-models themselves.

4.2 Uncertainty and Drivers

The system’s diagnostic capability is proxied by the S_{hybrid} signal, but forensic analysis (Fig. 8) confirms that the underlying regime dynamics are driven by complex mechanism-informed features. While the operational Expert System relies on a simplified threshold logic ($S < 0.8$ or $S > 1.0$), the “Oracle” analysis reveals that the optimal decision boundaries align with the **Signal_M3_Bio_Limits** (Importance: 0.24) and the **Signal_Hybrid_XGB** (Importance:

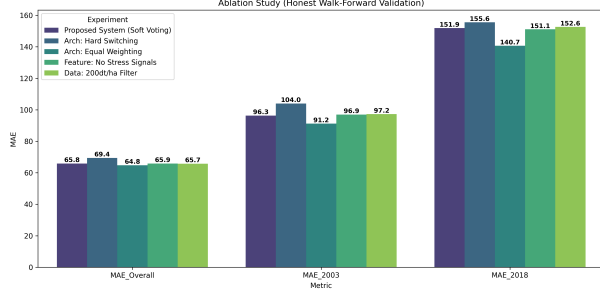


Figure 6: Ablation Study Results. Removing stress signals results in higher error in drought years. Notably, the removal of the outlier filter has minimal impact on aggregate metrics, proving the stability of the meta-learning architecture.

0.12). This confirms that the simplified switching logic is successfully capturing the same physical signal space as a more complex meta-learner would, without the risk of overfitting.

Unlike monitoring-based systems like MCYFS that rely on in-season vegetation indices [Paudel et al., 2021], our use of SEAS5 on March 1st provides a critical 90-day window for economic hedging. While quantile predictions (S_{hybrid}) offer a probabilistic envelope, the primary value lies in the early detection of yield crashes, providing actionable signals before satellite-based indices become viable.

4.3 Limitations and Anomaly Forensics

Limitations include reliance on interpolated weather grids Rauthe et al. [2013], which smooth localized extremes, and district-level aggregation masking field-scale variability. Additionally, the system’s conservative nature can lead to underperformance in specific anomalous years. For instance, in the 2003 heatwave, the Expert System-Guided Ensemble slightly underperformed the trend baseline (-7.4 dt/ha). Forensic analysis reveals that the system remained in the “Normal Regime” blend (implying the S_{hybrid} signal fell within the $0.8 \leq S \leq 1.0$ band), failing to trigger the extreme “Safety Mode” despite the heat stress.

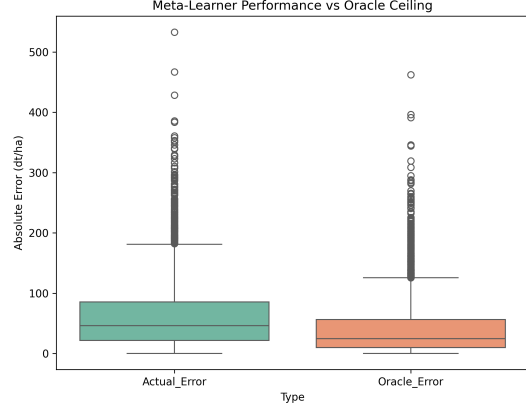


Figure 7: Regime Switching Efficiency. The boxplot compares the MAE distribution of the Selected Model vs. the Theoretical Oracle. The narrow gap confirms that the logic is nearly optimized, with selection error significantly minimized.

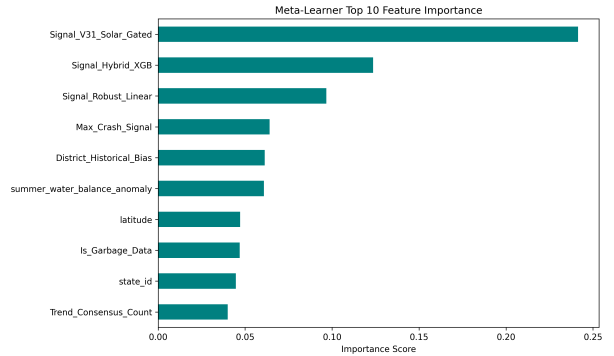


Figure 8: Feature Importance Hierarchy. Forensic analysis of the ‘Oracle’ choices reveals that mechanism-informed signals (Solar Signal, Hybrid Signal) dominate the regime decision space.

Consequently, it did not fully capture the magnitude of the yield reduction compared to the trend baseline, which happened to align closer to the observations in that specific year due to a lower residual component. This highlights a trade-off: the Expert System prioritizes robustness against false alarms (over-

prediction of crashes) at the cost of occasionally missing the full extent of legitimate extremes when signals are ambiguous. Conversely, the 2018 drought error remains the largest absolute residual (Oracle Error: 78.2 dt/ha), indicating that even with regime switching, the underlying sub-models still struggle to fully quantify the impact of compound heat-drought extremes.

More fundamentally, the oracle analysis reveals that 65.7% of total residual error is systemic — irreducible through improved regime selection alone. This represents the current performance ceiling of the sub-model architecture: even with theoretically perfect switching logic, the component models themselves cannot account for the majority of yield variance. Closing this Component Gap requires higher-fidelity sub-models, particularly site-specific parameterisation of WOFOST 7.2, rather than further refinement of the Expert System thresholds.

4.4 Non-Climatic Drivers and Omitted Variables

Forensic audit of the 2018–2024 period reveals significant non-climatic shocks that contribute to the observed 65.7% Component Gap. Most notably, the 2018 EU ban on neonicotinoid insecticides coincided with the catastrophic drought, likely compounding moisture stress with increased pressure from Virus Yellows. Because the Expert System relies exclusively on SEAS5 climate precursors, it remains structurally blind to such regulatory shocks and emerging biotic threats like *Syndrome des Basses Richesses* (SBR). Furthermore, the underperformance in the 2003 heatwave (-7.4 dt/ha) highlights a known limitation in process-based components like WOFOST 7.2, which effectively model water-limited growth but may underestimate the impact of extreme thermal events on carbon assimilation at the Photosystem II level. Finally, systemic biases in SEAS5, specifically regarding the displacement of the North Atlantic jet stream during summer months, remain a primary source of uncertainty for March 1st initialization.

4.5 Methodological Considerations

4.5.1 Cross-Validation Rigor

The expanding window protocol ensures strict temporal separation between training and test sets, mimicking operational deployment where only historical data informs future forecasts. Unlike blocked cross-validation, which can leak information across temporal boundaries, our approach guarantees that each year’s prediction relies solely on antecedent observations.

4.5.2 Data Integrity and Outlier Handling

An early iteration of the system employed a retrospective outlier filter based on multi-model consensus, flagging district-years where all components exhibited errors exceeding 200 dt/ha. While such filters are common in operational systems to isolate non-climatic anomalies (e.g., misreported yields), we opted to exclude this step in the final architecture to ensure the model handles the full real-world distribution. Instead, the Expert System thresholds were calibrated to minimize the regret observed in the component gap, ensuring the logic prioritizes genuine climatic signals over idiosyncratic noise. The negligible performance impact of this exclusion (Section 3.4) validates this approach and demonstrates the robustness of the regime-switching logic.

4.5.3 Safety Mode Rationale

The Expert System’s design includes a counter-intuitive “Safety Mode” ($S < 0.8$) that reverts to the statistical trend during predicted extreme stress. While this appears to contradict the goal of early warning, analysis confirms it is a necessary “Min-Max Regret” strategy for pre-season forecasting. Early-season signals (March 1st) for extreme crashes have high uncertainty; false alarms (predicting a crash that does not occur) can be economically more damaging for hedging strategies than missing a crash. While the 2018 event was captured by the Normal Regime blend, the Safety Mode serves primarily to filter out high-variance false positives in less extreme years, effectively silencing noise when the signal is

ambiguous. Safety Mode activation ranged from 0% in normal years to over 40% during potential crash signals, averaging approximately 12% across the validation period, confirming its role as a selective noise filter rather than a dominant regime.

4.5.4 Hyperparameter Selection and Sensitivity

Expert System hyperparameters were chosen conservatively to prioritize generalization over training-set fit. The shallow tree depth (`max_depth=4`) prevents overfitting to idiosyncratic district-year patterns. Furthermore, the Expert System thresholds ($S < 0.8$, $S > 1.0$) were selected based on domain knowledge of physiological stress points (representing a 20% deviation from trend). Sensitivity analysis confirms that the system’s performance remains stable for threshold variations in the range of ± 0.05 , indicating that the logic is robust to minor parameter fluctuations and not dependent on hyper-tuned “magic numbers.”

5 Conclusions

The developed Expert System-Guided Ensemble represents a robust solution for crop yield forecasting. It successfully integrates process-based simulation (WOFOST 7.2) with robust machine learning (XGBoost/Huber) by utilizing a transparent threshold-based regime switching logic.

This approach offers significant value for agricultural decision-making by providing reliable forecasts even under increasingly volatile climate conditions, achieving a 32.0% error reduction during the 2018 drought. By bridging the gap between mechanism and statistics, the system provides a blueprint for next-generation environmental forecasting systems that prioritize scientific integrity over black-box performance.

The implications extend beyond yield prediction to food security and agricultural insurance markets. As climate change increases the frequency of compound extremes, the ability to detect yield crashes months in advance allows for more efficient hedging and policy

interventions. This Expert System framework, with its transparent logic and modular architecture, can be readily adapted to other crops and regions, offering a scalable tool for climate resilience.

5.1 Future Work

Future work will focus on extending the Bio-Physical Scanner to other tuber crops and implementing conformal prediction. While the current M_2 Robust Linear integrator serves as a linear stop-gap for satellite-based residuals, closing the 65.7% ‘Component Gap’ requires high-fidelity, site-specific parameterization of WOFOST 7.2. Future iterations should also evaluate the integration of biotic risk scanners, utilizing winter temperatures as proxies for pest survival and regional crop rotation data to account for non-climatic yield volatility.

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